

VALVULAR HEART DISEASE — Clinical Revision Table

Bold = high-yield. RHD = commonest in developing world; degenerative in industrialized.

RHEUMATIC FEVER				
Item	Key points			
Cause	Immune response to group A strep ; cross-reacts cardiac myosin. Aschoff nodules pathognomonic			
Jones criteria	2 major OR 1 major + 2 minor + preceding strep. Major: carditis, arthritis, chorea, erythema marginatum, SC nodules			
Carditis	Pancarditis; soft systolic (MR) common; Carey Coombs (mid-diastolic valvulitis); AR 50%			
Arthritis	Migratory, asymmetric, large joints ; commonest major			
Chorea	Sydenham's (late ≥3mo, females); ¼ → chronic valve disease			
Skin	Erythema marginatum (trunk, never face); SC nodules (extensor, confirm dx)			
Treatment	Bed rest; benzathine penicillin + long-term prophylaxis; aspirin (arthritis), steroids (carditis)			
LEFT-SIDED STENOSIS & REGURGITATION				
Lesion	Cause	Murmur / signs	Symptoms	Management
Mitral stenosis	Almost always rheumatic	Loud S1 (tapping apex), opening snap, mid-diastolic murmur + pre-systolic accentuation , malar flush	Dyspnoea, fatigue, AF , haemoptysis, emboli	Anticoagulate + rate control; balloon valvuloplasty (TOC); MVR if calcified
Mitral regurgitation	RHD, MVP, degenerative, ischaemic	Pansystolic murmur → axilla, soft S1. MVP = mid-systolic click	Often few (chronic); acute = sudden deterioration	Diuretics/vasodilator; MV repair = TOC for severe
Aortic stenosis	Bicuspid (young), calcific (old), rheumatic	Harsh ejection systolic → neck, soft S2, S4; slow-rising pulse (parvus et tardus) , thrill	Angina, SOB, syncope (exertional); die 3–5yr untreated	Symptomatic severe → AVR / TAVI ; balloon only congenital
Aortic regurgitation	RHD, root dilation, bicuspid, IE	Early diastolic murmur (LSE, expiration); Austin Flint ; collapsing pulse, wide pulse pressure	Palpitation, SOB, PND, angina	AVR if symptomatic or EDD ≥55mm ; root replacement if Marfan
RIGHT-SIDED & PULMONARY				
Lesion	Key points			
Tricuspid stenosis	Rheumatic (+ MS); systemic congestion (ascites, hepatomegaly, oedema, prominent a wave, Kussmaul); severe ≤1.0 cm²; diuretics/valvotomy			
Tricuspid regurgitation	Usually functional (RV dilatation/right HF); giant v wave, pulsatile liver , pansystolic LSE; improves with HF treatment; annuloplasty if MVR			
Pulmonary stenosis	Almost always congenital (± TOF, Noonan); ejection systolic + click → left shoulder; severe = fixed split S2; balloon valvuloplasty if gradient >50			
Pulmonary regurgitation	Rare; PHT/PA dilatation; Graham Steell murmur (mimics AR); trivial PR normal			
PROSTHETIC VALVES				
Type	Key points			
Mechanical	Durable; lifelong anticoagulation (thromboembolism/thrombosis); clicks; strut fracture			
Biological	No anticoagulation needed but less durable (degenerate 7–10yr, worse mitral); best >65 / aortic			
TAVI	Bioprosthetic via femoral; native valve compressed; no sternotomy; high-risk/elderly. Stroke 2%, heart block needing pacemaker 5–15%			